

# Supplementary Material for: How Much of Deep Joint Source–Channel Coding’s Advantage Survives a Fair Comparison? Evidence from Image Classification

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## S1. FULL ALGORITHM PSEUDO-CODE

This section translates the three systems described in the main paper into pseudo-code. It is complementary to the paper and aims to make the experiments reproducible. Every design choice was made on the validation split; all results in the paper and in Section S5 come from the single preregistered evaluation on the 3925-image test split. The code is at <https://github.com/nxck2005/capstone>, and Table S1 lists where each algorithm is implemented.

Notation follows the main paper.  $k$  is the number of complex channel uses per image and  $c$  the number of complex latent channels, with  $c = 8$  at  $r = 1/6$  ( $k = 12,800$ ) and  $c = 2$  at  $r = 1/24$  ( $k = 3200$ ).  $s$  is the SNR  $E_s/N_0$  in dB and GN is group normalization with 8 groups. All experiment constants are read from a single parameter file, `spec/params.generated.yaml`; none of the code hard-codes an experiment setting.

### A. Deep JSCC

Algorithm S1 gives the DJSCC forward pass. Every system draws its channel noise from `KeyedNoise`( $\nu, k$ ), which seeds a counter-based (Philox) generator from a noise identity  $\nu$  alone, draws  $\mathbf{a}, \mathbf{b} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_k)$ , and returns  $(\mathbf{a} + i\mathbf{b})/\sqrt{2}$ . At evaluation  $\nu$  is a hash of the dataset version, the split manifest, the image’s stable identifier, the bandwidth ratio, the SNR, the channel seed, and  $k$ . It does not depend on the system, so every system at the same bandwidth ratio sees the same noise vector for the same image and SNR, regardless of batching or evaluation order. During training  $\nu$  also includes the training seed and the epoch, so each image receives fresh noise every epoch.

Algorithm S2 is one training epoch. The fixed-SNR and SNR-randomized models differ only in how  $s_n$  is chosen. The PAPR-capped model is trained with the same algorithm and a cap  $A = 3$  dB in Algorithm S1.

Algorithm S3 clips symbols whose scaled power would exceed the cap and scales all others up by a common factor until the mean power returns to one, without changing any phase. A solution exists for any cap of at least 0 dB. After projection the code asserts that the mean power is within  $10^{-6}$  of one and that the PAPR is within  $10^{-4}$  dB of the cap.

## ALGORITHM S1: DJSCC forward pass

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**input:** image  $\mathbf{x} \in [0, 1]^{3 \times 160 \times 160}$ , SNR  $s$ , noise identity  $\nu$ , optional PAPR cap  $A$   
**output:** symbols  $\mathbf{z} \in \mathbb{C}^k$ , reconstruction  $\hat{\mathbf{x}}$ , class scores  $\mathbf{s}$

*Encoder*  
**var**  $\mathbf{h} \leftarrow (\mathbf{x} - \boldsymbol{\mu})/\boldsymbol{\sigma}$   
 $\mathbf{h} \leftarrow \text{PReLU}(\text{GN}(\text{Conv}_{5 \times 5/2}^{64}(\mathbf{h})))$  64 × 80 × 80  
 $\mathbf{h} \leftarrow \text{PReLU}(\text{GN}(\text{Conv}_{5 \times 5/2}^{128}(\mathbf{h})))$  128 × 40 × 40  
**for**  $b \leftarrow 1$  **to** 2 **do**  
    **var**  $\mathbf{t} \leftarrow \text{PReLU}(\text{GN}(\text{Conv}_{3 \times 3}(\mathbf{h})))$   
     $\mathbf{h} \leftarrow \text{PReLU}(\mathbf{h} + \text{GN}(\text{Conv}_{3 \times 3}(\mathbf{t})))$  *residual block*  
**end**  
**var**  $\mathbf{p} \leftarrow \text{Conv}_{3 \times 3}^{2c}(\mathbf{h})$  2c × 40 × 40  
*Pack channel pairs into complex symbols, channel-major*  
**var**  $\mathbf{z} \leftarrow \text{flatten}(\mathbf{p}_{0::2} + i\mathbf{p}_{1::2})$   $k = 1600c$   
 $\mathbf{z} \leftarrow \mathbf{z} / \sqrt{\frac{1}{k} \sum_i |z_i|^2}$  *unit power per image*  
**if**  $A$  is set **then**  $\mathbf{z} \leftarrow \text{PeakProject}(\mathbf{z}, A)$   
*Channel*  
**var**  $\mathbf{r} \leftarrow \mathbf{z} + 10^{-s/20} \cdot \text{KeyedNoise}(\nu, k)$   
*Decoder*  
**var**  $\mathbf{g} \leftarrow \text{PReLU}(\text{GN}(\text{Conv}_{3 \times 3}^{128}(\text{unpack}(\mathbf{r}))))$   
**for**  $b \leftarrow 1$  **to** 2 **do** apply a residual block to  $\mathbf{g}$  as above  
**var**  $\hat{\mathbf{x}} \leftarrow \sigma(\text{Conv}_{4 \times 4/2}^3(\text{PReLU}(\text{GN}(\text{Conv}_{4 \times 4/2}^{64}(\mathbf{g}))))$   
**var**  $\mathbf{s} \leftarrow \mathbf{W} \text{GAP}(\mathbf{g}) + \mathbf{b}$  *one linear layer*  
**return**  $\mathbf{z}, \hat{\mathbf{x}}, \mathbf{s}$

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### B. Separated digital baseline

Algorithm S4 finds the transport-block payload for one operating point, and Algorithm S5 runs one image through the chain. A delivered image is classified by the fine-tuned ResNet-18 (the clean one is kept as a secondary stream). Every other outcome is scored with the outage rule, which predicts class 0. It is correct on exactly 100 of the 1000 class-balanced validation images, the value used in selection, and on 387 of the 3925 test images (9.9%).

Algorithm S6 chooses the operating point for each SNR. It was run twice. The first pass used the clean classifier’s codec accuracy. The second, which produced the operating points used in the paper, used the fine-tuned classifier’s accuracy and left everything else unchanged. The selected point is then re-run image by image through Algorithm S5, and that realized outcome is what the paper reports.

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**ALGORITHM S2:** One DJSCC training epoch

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**input:** model, epoch  $e$  of  $E = 100$ , weight  $\lambda = 3$ , base rate  $\eta_0 = 10^{-3}$   
 $\eta_e \leftarrow \frac{1}{2}\eta_0 (1 + \cos(\pi e/(E-1)))$  *set at epoch start*  
order the training set by a permutation keyed on (seed,  $e$ )  
**foreach** batch  $\{(\mathbf{x}_n, y_n)\}$  of 32 images **do**  
     $\mathbf{x}_n \leftarrow$  random resized crop and horizontal flip of  $\mathbf{x}_n$   
     $s_n \leftarrow 7$  dB, **or**  $s_n \sim \text{Uniform}\{1, 4, 7, 13, 19\}$  dB keyed per image and epoch  
     $(\mathbf{z}_n, \hat{\mathbf{x}}_n, s_n) \leftarrow$  Algorithm S1 under float16 autocast  
     $\mathcal{L} \leftarrow \text{mean}_n [\text{CE}(s_n, y_n) + \lambda \text{MSE}(\hat{\mathbf{x}}_n, \mathbf{x}_n)]$   
    scaled backward pass; Adam step at rate  $\eta_e$ , skipped if a gradient overflows  
**end**  
evaluate validation top-1 accuracy at 7 dB with keyed noise  
keep the checkpoint if it is the best so far (earliest epoch on ties)

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**ALGORITHM S3:** Peak projection

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**function** PeakProject ( $\mathbf{z} \in \mathbb{C}^k, A$ )  
    **var**  $P_i \leftarrow |z_i|^2$ ;  $C \leftarrow 10^{A/10}$   
    **var**  $\ell \leftarrow 0$ ;  $u \leftarrow 1$   
    *Find an upper bracket for the power scale*  
    **for**  $t \leftarrow 1$  **to** 64 **do**  
        **if**  $\frac{1}{k} \sum_i \min(uP_i, C) < 1$  **then**  $u \leftarrow 2u$   
    **end**  
    *Bisect on the common power scale*  
    **for**  $t \leftarrow 1$  **to** 64 **do**  
        **var**  $m \leftarrow (\ell + u)/2$   
        **if**  $\frac{1}{k} \sum_i \min(mP_i, C) < 1$  **then**  $\ell \leftarrow m$  **else**  $u \leftarrow m$   
    **end**  
     *$u$  is computed from detached powers*  
    **return**  $z_i \sqrt{\min(uP_i, C)/P_i}$  for all  $i$

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### C. Task-aware digital control

Algorithm S7 covers the task-aware digital system. Because the digital link either delivers the indices exactly or fails, the model is trained without a channel: its received features equal the transmitted ones whenever a prediction is made. This is also why its accuracy is constant wherever packets arrive: 82.0% on validation and 81.1% on the test split, averaged over three seed pairs (Table S8). The label-transmission system uses the same trained model: the transmitter computes  $\hat{y}$  locally, writes it into the low four bits of one byte, zero-pads that byte to the same payload  $A$ , and sends it through the same packet, so it shares the same decoding threshold.

The rate-1/5 variant runs Algorithm S7 unchanged with the  $D = 1024$  model from the same selection (Table S7), whose 2048-bit raw payload is sent in one BPSK transport block at LDPC rate 1/5 on base graph 2, with a 2544-bit payload, at every SNR. The installed decoder builds an empty parity-check matrix when rate matching prunes no variable nodes, which happens at exactly this rate; for that case only, the decoder is rebuilt without pruning.

## S2. CORE CODE

Listings S1–S4 are the actual implementations of the most important components, with argument validation and error handling removed for length. Names and arithmetic are unchanged.

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**ALGORITHM S4:** Transport-block payload

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**input:** channel uses  $k$ , bits per symbol  $Q_m$ , nominal LDPC rate  $R$   
**output:** payload size  $A$  in bits, or infeasible  
**var**  $G \leftarrow kQ_m$ ;  $N \leftarrow \lfloor GR \rfloor$   
*Largest byte-aligned payload whose packetization fits*  
**for**  $A \leftarrow 8\lfloor N/8 \rfloor$  **down to** 8 **step** 8 **do**  
    segment  $A$  per TS 38.212: TB CRC, base graph, code blocks with CB CRC,  $Z, K'$   
    distribute  $G$  over the code blocks as budgets  $E_1, \dots, E_C$   
    **if**  $K'/\min_r E_r \leq 0.95$  **and**  $A + |\text{CRC}_{\text{TB}}| \leq N$  **then break**  
**end**  
**if no**  $A$  qualified **then return** infeasible  
*Respect the base graph's minimum code rate*  
**while**  $K'/\max_r E_r < R_{\min}(\text{BG})$  **do**  $A \leftarrow A + 8$  and re-plan  
**return**  $A$

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**ALGORITHM S5:** Classical chain for one image

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**input:** image  $\mathbf{x}$ , encode size  $a$ , LDPC rate  $R$ , modulation, SNR  $s$ , noise identity  $\nu$   
**output:** structural infeasibility, codec infeasibility, decode failure, or delivered  $\hat{\mathbf{x}}$   
 $A \leftarrow$  Algorithm S4  
**if infeasible then return** structural infeasibility  
*Source coding*  
 $\mathbf{x}_a \leftarrow$  bilinear downsample of  $\mathbf{x}$  to  $a \times a$   
bisect the JPEG 2000 compression ratio for the largest codestream of at most  $A/8$  bytes  
**if no codestream fits then return** codec infeasibility  
zero-pad the codestream to exactly  $A/8$  bytes  
*Channel coding and modulation*  
TB CRC; segmentation and CB CRC; LDPC encoding; rate matching with filler  
bit interleaving; Gray-mapped BPSK, QPSK, or 16-QAM at unit average energy  
add  $10^{-s/20} \cdot \text{KeyedNoise}(\nu, k)$   
max-log LLRs; offset min-sum decoding (offset 0.5, 50 iterations)  
**if any CRC fails then return** decode failure  
*Source decoding*  
strip filler; decode the codestream; bicubic upsample to  $160 \times 160$   
**return** delivered,  $\hat{\mathbf{x}}$

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Listing S1. DJSCC residual block, complex packing, and encoder forward pass (src/models/djscc.py).

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```

class ResidualBlock(nn.Module):
    def __init__(self, width):
        super().__init__()
        self.conv1 = nn.Conv2d(width, width, kernel_size=3, padding=1)
        self.norm1 = nn.GroupNorm(8, width)
        self.activation1 = nn.PReLU(width)
        self.conv2 = nn.Conv2d(width, width, kernel_size=3, padding=1)
        self.norm2 = nn.GroupNorm(8, width)
        self.activation2 = nn.PReLU(width)

    def forward(self, inputs):
        residual = self.activation1(self.norm1(self.conv1(inputs)))
        residual = self.norm2(self.conv2(residual))
        return self.activation2(inputs + residual)

def pack_complex_symbols(projection):
    """Pack adjacent real/imaginary channels, then flatten C, row, column."""
    batch, doubled_channels, height, width = projection.shape
    paired = projection.reshape(batch, doubled_channels // 2, 2, height, width)
    if paired.dtype in (torch.float16, torch.bfloat16):
        paired = paired.float() # keep the channel interface at complex64 under AMP
    symbols = torch.complex(paired[:, :, 0], paired[:, :, 1])
    return symbols.flatten(start_dim=1)

class DJSCCEncoder(nn.Module):
    def forward(self, inputs):
        features = self.residual(self.body_entry(self.stem(self.input_normalisation(inputs))))
        symbols = pack_complex_symbols(self.projection(features))
        symbols = normalize_unit_average_power(symbols)
        if self.peak_constraint is not None:
            symbols = self.peak_constraint(symbols)
        return symbols

```

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**ALGORITHM S6:** Per-SNR operating-point selection

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**input:** SNR  $s$ ; candidates  $\mathcal{C}$  of (encode size,  $R$ , modulation);  
 measured BLER curves; measured codec accuracy  $A_{\text{codec}}(c)$ ;  
 outage accuracy  $A_{\text{out}} = 0.100$

**output:** operating point  $c^*$

**foreach** feasible  $c \in \mathcal{C}$  **do**

**foreach** code block  $r$  of  $c$ 's packet **do**

$\text{BLER}_r(s) \leftarrow$  measured curve, linear in  $\log_{10}$  BLER  
         between measured SNRs

**if**  $s$  is outside the measured range **then** exclude  $c$

**end**

$P_s(c) \leftarrow \prod_r (1 - \text{BLER}_r(s))$

$E(c) \leftarrow P_s(c) A_{\text{codec}}(c) + (1 - P_s(c)) A_{\text{out}}$

**end**

Exact ties: higher  $P_s$ , lower  $Q_m$ , lower  $R$ , larger image, then identifier

**return**  $\arg \max_c E(c)$

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**ALGORITHM S7:** Task-aware digital system

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**input:**  $D = 2048$ ,  $b = 2$  bits,  $L = 4$  levels, step  $\Delta = 2/(L - 1)$

*Encoder*

**var**  $\mathbf{h} \leftarrow$  DJSCC encoder trunk (Algorithm S1, up to the residual blocks)

**var**  $\mathbf{u} \leftarrow \text{AvgPool}_{8 \times 8}(\text{Conv}_{3 \times 3}^{D/64}(\mathbf{h}))$   $32 \times 8 \times 8$

**var**  $\mathbf{q} \leftarrow \text{clamp}(\text{round}((\tanh \mathbf{u} + 1)/\Delta), 0, L - 1)$

*Training: no channel, straight-through estimator*

$\tilde{\mathbf{u}} \leftarrow \tanh \mathbf{u} + \text{sg}(\mathbf{q}\Delta - 1 - \tanh \mathbf{u})$

$\mathcal{L} \leftarrow \text{CE}(\mathbf{W} \text{ GAP}(\tilde{\mathbf{u}}) + \mathbf{b}, \mathbf{y})$

*Transmission*

**var**  $\mathbf{m}_{\text{rc}} \leftarrow$  range-code  $\mathbf{q}$  with a static model fitted on training indices

**var**  $\mathbf{m}_{\text{raw}} \leftarrow \mathbf{q}$  as  $bD = 4096$  raw bits

**if**  $|\mathbf{m}_{\text{rc}}| \leq |\mathbf{m}_{\text{raw}}|$  **then**  $\mathbf{m} \leftarrow [0, \mathbf{m}_{\text{rc}}]$  **else**  $\mathbf{m} \leftarrow [1, \mathbf{m}_{\text{raw}}]$

send  $\mathbf{m}$ , zero-padded to  $A$ , through the channel-coding steps of Algorithm S5 (BPSK, rate 1/3)

*Reception*

**if** a CRC fails **then** predict the outage class **else**

$\hat{\mathbf{y}} \leftarrow \arg \max(\mathbf{W} \text{ GAP}(\mathbf{q}\Delta - 1) + \mathbf{b})$

---

Listing S2. Power normalization, symbol-domain PAPR, and the peak projection (src/channels/power.py).

---

```

def normalize_unit_average_power(symbols):
    """Normalise every image independently to mean_k  $|x|^2 = 1$ ."""
    powers = symbols.abs().square()
    return symbols * torch.rsqrt(powers.mean(dim=1)).unsqueeze(1)

def symbol_papr_db(symbols):
    """Symbol-domain PAPR in dB per image (not oversampled waveform PAPR)."""
    powers = symbols.abs().square()
    return 10.0 * torch.log10(powers.amax(dim=1) / powers.mean(dim=1))

class PeakPowerConstraint(nn.Module):
    def forward(self, symbols):
        powers = symbols.abs().square()
        detached = powers.detach().to(torch.float64)
        cap = torch.as_tensor(self.max_power, dtype=torch.float64, device=symbols.device)
        low = torch.zeros((symbols.shape[0], 1), dtype=torch.float64, device=symbols.device)
        high = torch.ones_like(low)
        for _ in range(64):
            # bracket
            achieved = torch.minimum(high * detached, cap).mean(dim=1, keepdim=True)
            high = torch.where(achieved < 1, high * 2, high)
        for _ in range(64):
            # bisection
            midpoint = (low + high) / 2
            achieved = torch.minimum(midpoint * detached, cap).mean(dim=1, keepdim=True)
            low = torch.where(achieved < 1, midpoint, low)
            high = torch.where(achieved >= 1, midpoint, high)
        scale = high.to(powers.dtype)
        projected_powers = torch.minimum(scale * powers, powers.new_tensor(self.max_power))
        support = powers > 0
        amplitude_scale = torch.zeros_like(projected_powers)
        amplitude_scale[support] = torch.sqrt(projected_powers[support] / powers[support])
        return symbols * amplitude_scale

```

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TABLE S1  
WHERE EACH COMPONENT IS IMPLEMENTED.

| Component                              | File                                                                   |
|----------------------------------------|------------------------------------------------------------------------|
| DJSCC model (Alg. S1)                  | src/models/djscc.py                                                    |
| Power, PAPR, peak projection (Alg. S3) | src/channels/power.py                                                  |
| Keyed AWGN                             | src/channels/awgn.py                                                   |
| Training objective                     | src/training/djscc_loss.py                                             |
| Training loops (Alg. S2)               | src/training/w8_final.py,<br>er2_randomized.py,<br>papr_constrained.py |
| Payload solver (Alg. S4)               | src/baseline/ldpc/transport.<br>py                                     |
| JPEG 2000 rate control                 | src/baseline/j2k.py                                                    |
| Classical chain (Alg. S5)              | src/baseline/classical/<br>pipeline.py                                 |
| Selection (Alg. S6)                    | src/baseline/classical/<br>composition.py                              |
| Task-aware model (Alg. S7)             | src/models/er9_digital.py                                              |
| Quantizer, message format              | src/evaluation/er9_protocol.<br>py                                     |
| Reference classifiers                  | src/models/reference_<br>classifier.py                                 |
| Per-image evaluation                   | src/evaluation/w10_backends.<br>py, w10_classical.py                   |
| Test campaign                          | src/evaluation/gl2_dispatch.<br>py, tools/run_gl2_campaign.<br>py      |
| Hypothesis tests                       | src/evaluation/gl2_analysis.<br>py                                     |

---

Listing S3. Keyed complex noise, the AWGN channel, and the training objective (src/channels/awgn.py, src/training/djscc\_loss.py).

```
def keyed_complex_noise(noise_ids, k, *, dtype=torch.complex64, device=None):
    """Unit-variance complex noise as a pure function of each noise_id."""
    rows = []
    for noise_id in noise_ids:
        components = keyed_standard_normal("channel_noise", {"noise_id": noise_id}, size=(2, k))
        rows.append((components[0] + 1j * components[1]) * math.sqrt(0.5))
    return torch.as_tensor(np.stack(rows), dtype=dtype, device=device)

class AWGN(nn.Module):
    """Complex AWGN with  $E|n|^2 = 1/\gamma$  after unit-power normalisation."""
    def forward(self, symbols, snr_db, *, unit_noise=None):
        snr = torch.as_tensor(snr_db, dtype=symbols.real.dtype, device=symbols.device)
        if snr.ndim == 0:
            snr = snr.expand(symbols.shape[0]) # scalar or one SNR per sample
        if unit_noise is None:
            unit_noise = torch.complex(torch.randn_like(symbols.real),
                                       torch.randn_like(symbols.real)) * math.sqrt(0.5)
        gamma = torch.pow(torch.as_tensor(10.0, device=symbols.device), snr / 10.0)
        return symbols + unit_noise * torch.rsqrt(gamma).unsqueeze(1)

class DJSCCObjective(nn.Module):
    """CE + lambda * MSE, with lambda retained as data, including zero."""
    def forward(self, output, targets, augmented_inputs):
        cross_entropy = F.cross_entropy(output.logits, targets)
        reconstruction_mse = F.mse_loss(output.reconstruction, augmented_inputs)
        total = cross_entropy + self.reconstruction_weight * reconstruction_mse
        return DJSCCLoss(total=total, cross_entropy=cross_entropy, reconstruction_mse=reconstruction_mse)
```

---

Listing S4. Operating-point composition and BLER interpolation (src/baseline/classical/composition.py), and the task-aware quantizer (src/evaluation/er9\_protocol.py).

```
def success_probability(block_blers):
    """P(TB success) = product over code blocks of (1 - BLER_r)."""
    probability = 1.0
    for bler in block_blers:
        probability *= 1.0 - bler
    return probability

def expected_accuracy(*, success_probability, codec_accuracy, outage_accuracy):
    # Both accuracy terms are measured counts on the validation split, never 1/n_classes.
    return success_probability * codec_accuracy.value + (1.0 - success_probability) * outage_accuracy.value

def _interpolate(curve, snr):
    """Linear in SNR against log10(BLER), strictly inside the measured span."""
    for point_snr, point_bler in zip(curve.snr_db, curve.bler):
        if point_snr == snr:
            return point_bler
    for i in range(len(curve.snr_db) - 1):
        low, high = curve.snr_db[i], curve.snr_db[i + 1]
        if low < snr < high:
            w = (snr - low) / (high - low)
            return 10 ** ((1 - w) * math.log10(curve.bler[i]) + w * math.log10(curve.bler[i + 1]))
    raise UncharacterizedBlerError(snr) # the candidate is excluded, not given a default

class UniformScalarQuantizer:
    def __init__(self, bits):
        self.levels = 2 ** bits
        self.step = 2.0 / (self.levels - 1) # fixed [-1, +1] range, no side information

    def quantise(self, features):
        scaled = (torch.tanh(features) + 1.0) / self.step
        return torch.round(scaled).clamp(0, self.levels - 1).to(torch.long)

    def dequantise(self, indices):
        return indices.to(torch.float32) * self.step - 1.0

    def straight_through(self, features):
        bounded = torch.tanh(features)
        indices = self.quantise(features)
        dequantised = self.dequantise(indices).to(features.dtype)
        return bounded + (dequantised - bounded).detach(), indices
```

---

### S3. ARCHITECTURE AND TRAINING SETTINGS

Table S2 lists every system variant evaluated on the test split, Table S3 gives the model sizes, and Table S4 the training and link settings.

*Implementation checks.* Table S5 summarizes the checks run before the test campaign. The clean ResNet-18 reaches 898/1000 on the validation split after 100 epochs, above its target of 0.88. A batch-32 profile of the DJSCC model at  $r = 1/2$  on an RTX 4060 Laptop GPU measured 1.004 GiB of peak reserved memory and 48.7 s per epoch, or about 1.35 h for 100 epochs.

The TS 38.212-derived LDPC core was checked against an independent reference implementation with  $K = 128$ ,  $N = 256$ , base graph 2, lifting size 22, rate  $1/2$ , offset min-sum decoding, and 5000 blocks per SNR point; across BPSK, QPSK, and 16-QAM the waterfall at BLER  $10^{-2}$  is within 0.0037 dB of the reference. Packetization was checked across 216 configurations, of which 215 are feasible and one is structurally infeasible under the configured budget. The BLER curves used for selection cover 3213 operating points in 153 curves at 5000 trials each; operating points without a measured curve are excluded from selection rather than given a default BLER. The classical validation sweep evaluated 288 structural configurations over all 1000 validation images: 264,000 of the 288,000 evaluations were delivered and 24,000 were codec-infeasible, with no decode or structural failures, and selection produced a valid operating point for all 378 requested SNR cells.

A design study on validation data, run before the final models were trained, estimated that at 80% power the smallest paired difference H4 could detect at a single SNR is 1.8 to 1.9 points above the digital cliff and about 4 points below it.

TABLE S2

SYSTEM VARIANTS EVALUATED ON THE TEST SPLIT. EVERY VARIANT USES THE SAME COMPLEX AWGN CHANNEL AND THE SAME  $k$  FOR ITS BANDWIDTH RATIO. SEEDS: NUMBER OF TRAINED (OR, FOR THE UNTRAINED DIGITAL CODECS, CHANNEL) SEED PAIRS EVALUATED.

| Variant                      | $r$  | Seeds | Classifier at receiver   |
|------------------------------|------|-------|--------------------------|
| DJSCC (7 dB training)        | 1/6  | 3     | Own task head            |
|                              | 1/24 | 1     | Own task head            |
| DJSCC, SNR-randomized        | 1/6  | 1     | Own task head            |
| DJSCC, PAPR-capped           | 1/6  | 1     | Own task head            |
| DJSCC reconstruction         | 1/6  | 1     | Clean ResNet-18          |
| JPEG 2000 + LDPC, adaptive   | 1/6  | 3     | Fine-tuned ResNet-18     |
|                              | 1/24 | 1     | Fine-tuned ResNet-18     |
| JPEG 2000 + LDPC, QPSK only  | 1/6  | 1     | Fine-tuned ResNet-18     |
| JPEG 2000 + LDPC, fixed      | 1/6  | 3     | Fine-tuned ResNet-18     |
| Baseline JPEG + LDPC         | 1/6  | 1     | Fine-tuned ResNet-18     |
| Task-aware digital           | 1/6  | 3     | Own task head            |
| Task-aware digital, rate 1/5 | 1/6  | 1     | Own task head            |
| Label transmission           | 1/6  | 1     | Transmitter's prediction |

TABLE S3  
MODEL SIZES (PARAMETERS).

| Model                      | Encoder | Decoder | Total      |
|----------------------------|---------|---------|------------|
| DJSCC, $r = 1/6$           | 820,688 | 746,509 | 1,567,197  |
| DJSCC, $r = 1/24$          | 806,852 | 732,685 | 1,539,537  |
| DJSCC, $r = 1/2$ (profile) | 857,584 | 783,373 | 1,640,957  |
| ResNet-18 classifier       | —       | —       | 11,181,642 |

TABLE S4  
TRAINING AND DIGITAL-LINK SETTINGS.

| Setting               | Value                                                                                                                                                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Optimizer             | Adam, $\beta = (0.9, 0.999)$ , $\epsilon = 10^{-8}$ , no weight decay                                                                                                                                                    |
| Learning rate         | $10^{-3}$ , cosine to 0 over 100 epochs, no warm-up                                                                                                                                                                      |
| Batch                 | 32 images, last partial batch kept                                                                                                                                                                                       |
| Augmentation          | random resized crop, horizontal flip                                                                                                                                                                                     |
| Precision             | float16 autocast, dynamic gradient scaling                                                                                                                                                                               |
| Training SNR          | 7 dB, or $\{1, 4, 7, 13, 19\}$ dB per image and epoch                                                                                                                                                                    |
| Loss                  | CE + $3 \cdot \text{MSE}$ (DJSCC); CE (task-aware digital)                                                                                                                                                               |
| Selection             | best validation top-1 at 7 dB, earliest on ties                                                                                                                                                                          |
| Seeds                 | three paired seeds per DJSCC ratio and for the task-aware model; one each for the randomized and PAPR-capped models; on the test split the main $r = 1/6$ systems are evaluated in all three pairs, the others in pair 0 |
| Fine-tuned classifier | from the clean ResNet-18, 20 epochs (5 warm-up) on 44,039 JPEG 2000-decoded training images                                                                                                                              |
| Channel code          | 5G NR LDPC per TS 38.212 V17.13.0; Sionna 2.0.1 encoder and decoder                                                                                                                                                      |
| LDPC rates            | 1/3, 1/2, 2/3, 5/6; rate 1/5 for the secondary task-aware variant only                                                                                                                                                   |
| Decoder               | offset min-sum, offset 0.5, 50 iterations, flooding, LLR clip 20                                                                                                                                                         |
| Modulation            | BPSK, QPSK, 16-QAM; Gray mapping; TS 38.212 bit interleaver; max-log demapping                                                                                                                                           |
| Source codec          | OpenJPEG 2.5.4: raw codestream, 9/7 wavelet, 6 levels, RPCL, $64 \times 64$ blocks                                                                                                                                       |
| Rate control          | bisection on compression ratio (16-byte tolerance, at most 24 encodes)                                                                                                                                                   |
| Resizing              | bilinear down, bicubic up                                                                                                                                                                                                |

TABLE S5  
IMPLEMENTATION CHECKS.

| Item             | Result                                                                              |
|------------------|-------------------------------------------------------------------------------------|
| Clean classifier | 0.898 validation top-1 (898/1000), ResNet-18 trained from scratch                   |
| DJSCC profile    | 1,640,957 parameters at $r = 1/2$ ; 1.004 GiB peak reserved VRAM at batch 32        |
| LDPC conformance | Waterfall offset $\leq 0.0037$ dB at BLER $10^{-2}$ against the reference decoder   |
| BLER campaign    | 153 curves; 3213 points; 5000 trials/point; 16,065,000 trials total                 |
| Classical sweep  | 288 configurations $\times$ 1000 images; 264,000 delivered, 24,000 codec-infeasible |

#### S4. VALIDATION SELECTION RECORDS

Table S6 records the choice of the reconstruction weight  $\lambda$  at  $r = 1/6$ . The rule admits every  $\lambda$  within one point of the  $\lambda = 0$  accuracy (floor 81.7%), keeps those whose mean reconstruction PSNR at 15 dB is at least 20 dB (falling back to 16 dB if none qualifies), and takes the smallest  $\lambda$  remaining. Only  $\lambda = 3$  reached 20 dB. Table S7 records the choice of the task-aware feature dimension, measured through the real digital chain at 7 dB.

TABLE S6  
CHOICE OF  $\lambda$ . ACCURACY IS VALIDATION TOP-1 AT 7 dB; PSNR IS AT 15 dB.

| $\lambda$ | Epoch | Acc. (%) | PSNR (dB) | $\geq 20$ dB |
|-----------|-------|----------|-----------|--------------|
| 0         | 84    | 82.7     | 9.7       |              |
| 0.1       | 75    | 83.5     | 17.6      |              |
| 0.3       | 71    | 82.2     | 18.7      |              |
| 1         | 67    | 82.8     | 19.9      |              |
| 3         | 92    | 83.2     | 21.2      | ✓            |

TABLE S7  
CHOICE OF THE TASK-AWARE FEATURE DIMENSION  $D$  ( $b = 2$ ).

| $D$  | Raw payload (bits) | Acc. (%) |
|------|--------------------|----------|
| 64   | 128                | 28.4     |
| 128  | 256                | 55.2     |
| 256  | 512                | 75.9     |
| 512  | 1024               | 79.3     |
| 1024 | 2048               | 81.8     |
| 2048 | 4096               | 82.0     |

#### S5. COMPLETE TEST RESULTS

Tables S8–S17 give every test measurement. Table I of the main paper is a subset of Table S8, and Table S14 gives the operating points of its adaptive JPEG 2000 baseline at  $r = 1/6$ . Table S9 shows the spread between seed pairs, Table S12 the paired differences plotted in Fig. 2 of the paper, and Table S13 the point statistics on which the H1 and H4 run rules operate. The operating points in Tables S14–S17 were selected on validation data; each was checked against the configuration actually run on the test split.

The symbol-domain PAPR over the 3925 test images, in the first seed pair, is: DJSCC at  $r = 1/6$ , mean 13.26 dB (13.26 to 14.79 dB across the three seed pairs) and maximum 24.65 dB over all three; DJSCC at  $r = 1/24$ , mean 12.32 dB and maximum 22.37 dB; SNR-randomized DJSCC, mean 13.25 dB and maximum 23.47 dB; PAPR-capped DJSCC, mean 3.000 dB and maximum 3.000002 dB; adaptive JPEG 2000 at  $r = 1/6$ , at most 2.61 dB mean and 3.34 dB maximum at any SNR.

TABLE S8

TEST TOP-1 ACCURACY (%) OF EVERY SYSTEM VARIANT AT EVERY MEASURED SNR (3925 IMAGES PER ENTRY, PRIMARY SCORER). “(3)” MARKS THE MEAN OVER THREE SEED PAIRS; ALL OTHER ROWS ARE THE FIRST SEED PAIR.

| System                 | SNR $E_s/N_0$ (dB) |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|------------------------|--------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|                        | -8                 | -7   | -6   | -5   | -4   | -3   | -2   | -1   | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 9    | 11   | 13   | 15   | 18   |
| DJSCC (3)              | 76.1               | 77.5 | 79.1 | 80.0 | 80.8 | 80.9 | 81.4 | 81.7 | 81.6 | 81.8 | 82.0 | 82.2 | 82.1 | 82.4 | 82.2 | 82.3 | 82.3 | 82.4 | 82.4 | 82.4 | 82.4 |
| DJSCC, SNR-rand.       | 77.6               | 78.8 | 80.0 | 80.5 | 81.1 | 81.9 | 82.1 | 82.1 | 82.6 | 82.6 | 82.6 | 83.0 | 82.9 | 82.9 | 82.9 | 82.6 | 82.8 | 82.9 | 83.0 | 83.0 | 83.0 |
| DJSCC, PAPR-capped     | 76.7               | 78.3 | 79.3 | 80.2 | 81.0 | 81.3 | 81.4 | 81.1 | 81.8 | 82.4 | 81.5 | 82.2 | 82.4 | 82.1 | 82.2 | 82.3 | 82.0 | 82.2 | 82.1 | 82.1 | 82.3 |
| DJSCC recon. + clean   | 44.1               | 48.5 | 52.5 | 56.8 | 60.3 | 62.8 | 65.3 | 68.8 | 70.9 | 72.6 | 74.1 | 74.6 | 75.8 | 76.3 | 76.7 | 77.3 | 77.4 | 78.1 | 77.7 | 78.0 | 78.0 |
| J2K adaptive (3)       | 9.9                | 9.9  | 9.9  | 9.9  | 81.6 | 81.6 | 83.2 | 85.9 | 85.9 | 85.9 | 86.2 | 86.2 | 86.2 | 86.2 | 86.2 | 86.5 | 86.5 | 84.3 | 86.6 | 86.6 | 86.6 |
| J2K QPSK only          | 9.9                | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 85.9 | 85.9 | 85.9 | 86.2 | 86.2 | 86.2 | 86.2 | 86.2 | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 |
| J2K fixed MCS (3)      | 9.9                | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 |
| JPEG adaptive          | 9.9                | 9.9  | 9.9  | 9.9  | 66.4 | 66.4 | 74.5 | 81.2 | 81.2 | 81.2 | 84.3 | 84.3 | 85.1 | 85.1 | 85.5 | 85.5 | 9.9  | 86.1 | 86.1 | 86.1 | 86.1 |
| Task-aware digital (3) | 9.9                | 9.9  | 9.9  | 9.9  | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 |
| Task-aware, rate 1/5   | 9.9                | 56.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 | 80.9 |
| Label transmission     | 9.9                | 9.9  | 9.9  | 9.9  | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 |
| DJSCC (1/24)           | 47.3               | 53.9 | 59.1 | 64.5 | 68.3 | 71.4 | 73.9 | 75.7 | 76.3 | 77.5 | 78.0 | 78.4 | 78.8 | 79.8 | 79.7 | 79.8 | 80.0 | 80.0 | 80.2 | 80.2 | 80.0 |
| J2K adaptive (1/24)    | 9.9                | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 28.9 | 67.9 | 68.0 | 72.6 | 79.1 | 79.1 | 81.6 | 81.6 | 83.9 | 84.9 | 81.5 | 85.9 | 86.3 | 86.3 | 86.3 |

TABLE S9

TEST TOP-1 ACCURACY (%) OF EACH SEED PAIR FOR THE SYSTEMS EVALUATED IN ALL THREE. FOR THE DIGITAL CODECS THE SEED PAIRS DIFFER ONLY IN THE CHANNEL NOISE.

|                            | SNR $E_s/N_0$ (dB) |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|----------------------------|--------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| System                     | -8                 | -7   | -6   | -5   | -4   | -3   | -2   | -1   | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 9    | 11   | 13   | 15   | 18   |
| DJSCC, seed 0              | 71.5               | 74.4 | 76.3 | 78.2 | 79.3 | 79.9 | 80.5 | 80.7 | 81.1 | 81.3 | 81.2 | 81.6 | 81.5 | 82.1 | 82.0 | 82.0 | 82.0 | 82.1 | 82.1 | 82.1 | 82.1 |
| DJSCC, seed 1              | 78.4               | 79.0 | 80.8 | 80.9 | 81.4 | 81.2 | 81.8 | 82.2 | 81.5 | 81.8 | 82.4 | 82.4 | 82.6 | 82.5 | 82.4 | 82.5 | 82.6 | 82.7 | 82.6 | 82.8 | 82.7 |
| DJSCC, seed 2              | 78.5               | 79.1 | 80.2 | 80.9 | 81.6 | 81.6 | 81.9 | 82.2 | 82.1 | 82.3 | 82.3 | 82.4 | 82.3 | 82.5 | 82.3 | 82.4 | 82.3 | 82.4 | 82.4 | 82.4 | 82.5 |
| J2K adaptive, seed 0       | 9.9                | 9.9  | 9.9  | 9.9  | 81.6 | 81.6 | 83.3 | 85.9 | 85.9 | 85.9 | 86.2 | 86.2 | 86.2 | 86.2 | 86.5 | 86.5 | 84.4 | 86.6 | 86.6 | 86.6 | 86.6 |
| J2K adaptive, seed 1       | 9.9                | 9.9  | 9.9  | 9.9  | 81.6 | 81.6 | 83.3 | 85.9 | 85.9 | 85.9 | 86.2 | 86.2 | 86.2 | 86.2 | 86.5 | 86.5 | 84.4 | 86.6 | 86.6 | 86.6 | 86.6 |
| J2K adaptive, seed 2       | 9.9                | 9.9  | 9.9  | 9.9  | 81.6 | 81.6 | 83.0 | 85.9 | 85.9 | 85.9 | 86.2 | 86.2 | 86.2 | 86.2 | 86.5 | 86.5 | 84.2 | 86.6 | 86.6 | 86.6 | 86.6 |
| J2K fixed MCS, seed 0      | 9.9                | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 |
| J2K fixed MCS, seed 1      | 9.9                | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 |
| J2K fixed MCS, seed 2      | 9.9                | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 |
| Task-aware digital, seed 0 | 9.9                | 9.9  | 9.9  | 9.9  | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 | 81.0 |
| Task-aware digital, seed 1 | 9.9                | 9.9  | 9.9  | 9.9  | 81.2 | 81.2 | 81.2 | 81.2 | 81.2 | 81.2 | 81.2 | 81.2 | 81.2 | 81.2 | 81.2 | 81.2 | 81.2 | 81.2 | 81.2 | 81.2 | 81.2 |
| Task-aware digital, seed 2 | 9.9                | 9.9  | 9.9  | 9.9  | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 | 81.1 |

TABLE S10

FRACTION OF PACKETS DELIVERED (%) ON THE TEST SPLIT FOR THE DIGITAL SYSTEMS. DJSCC VARIANTS ALWAYS PRODUCE AN OUTPUT AND ARE OMITTED.

|                        | SNR $E_s/N_0$ (dB) |      |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|------------------------|--------------------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| System                 | -8                 | -7   | -6    | -5    | -4    | -3    | -2    | -1    | 0     | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 9     | 11    | 13    | 15    | 18    |
| J2K adaptive (3)       | 0.0                | 0.0  | 0.0   | 0.0   | 100.0 | 100.0 | 97.6  | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 96.6  | 100.0 | 100.0 | 100.0 | 100.0 |
| J2K QPSK only          | 0.0                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| J2K fixed MCS (3)      | 0.0                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| JPEG adaptive          | 0.0                | 0.0  | 0.0   | 0.0   | 99.7  | 99.7  | 97.8  | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 0.0   | 100.0 | 100.0 | 100.0 | 100.0 |
| Task-aware digital (3) | 0.0                | 0.0  | 0.0   | 0.0   | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| Task-aware, rate 1/5   | 0.0                | 66.4 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| Label transmission     | 0.0                | 0.0  | 0.0   | 0.0   | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| J2K adaptive (1/24)    | 0.0                | 0.0  | 0.0   | 0.0   | 0.0   | 0.0   | 82.0  | 99.9  | 100.0 | 91.2  | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 94.5  | 100.0 | 100.0 | 100.0 | 100.0 |

TABLE S11

TEST ACCURACY (%) OF THE SAME DIGITAL OUTPUTS SCORED BY THE FINE-TUNED AND THE CLEAN RESNET-18.

|                                 | SNR $E_s/N_0$ (dB) |     |     |     |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|---------------------------------|--------------------|-----|-----|-----|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| System                          | -8                 | -7  | -6  | -5  | -4   | -3   | -2   | -1   | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 9    | 11   | 13   | 15   | 18   |
| J2K adaptive (3), fine-tuned    | 9.9                | 9.9 | 9.9 | 9.9 | 81.6 | 81.6 | 83.2 | 85.9 | 85.9 | 85.9 | 86.2 | 86.2 | 86.2 | 86.2 | 86.5 | 86.5 | 84.3 | 86.6 | 86.6 | 86.6 | 86.6 |
| J2K adaptive (3), clean         | 9.9                | 9.9 | 9.9 | 9.9 | 59.2 | 59.2 | 78.2 | 84.4 | 84.4 | 84.4 | 86.4 | 86.4 | 86.4 | 86.4 | 87.0 | 87.5 | 85.2 | 87.6 | 87.6 | 87.6 | 87.6 |
| J2K QPSK only, fine-tuned       | 9.9                | 9.9 | 9.9 | 9.9 | 9.9  | 9.9  | 9.9  | 85.9 | 85.9 | 85.9 | 86.2 | 86.2 | 86.2 | 86.2 | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 |
| J2K QPSK only, clean            | 9.9                | 9.9 | 9.9 | 9.9 | 9.9  | 9.9  | 9.9  | 84.4 | 84.4 | 84.4 | 86.4 | 86.4 | 86.4 | 86.4 | 87.0 | 87.0 | 87.0 | 87.0 | 87.0 | 87.0 | 87.0 |
| J2K fixed MCS (3), fine-tuned   | 9.9                | 9.9 | 9.9 | 9.9 | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 | 86.5 |
| J2K fixed MCS (3), clean        | 9.9                | 9.9 | 9.9 | 9.9 | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 9.9  | 87.5 | 87.5 | 87.5 | 87.5 | 87.5 | 87.5 |
| JPEG adaptive, fine-tuned       | 9.9                | 9.9 | 9.9 | 9.9 | 66.4 | 66.4 | 74.5 | 81.2 | 81.2 | 81.2 | 84.3 | 84.3 | 85.1 | 85.1 | 85.5 | 85.5 | 9.9  | 86.1 | 86.1 | 86.1 | 86.1 |
| JPEG adaptive, clean            | 9.9                | 9.9 | 9.9 | 9.9 | 45.0 | 45.0 | 60.5 | 70.6 | 70.7 | 70.7 | 83.5 | 83.5 | 85.6 | 85.6 | 86.6 | 86.6 | 9.9  | 87.0 | 87.0 | 87.0 | 87.0 |
| J2K adaptive (1/24), fine-tuned | 9.9                | 9.9 | 9.9 | 9.9 | 9.9  | 9.9  | 28.9 | 67.9 | 68.0 | 72.6 | 79.1 | 79.1 | 81.6 | 81.6 | 83.9 | 84.9 | 81.5 | 85.9 | 86.3 | 86.3 | 86.3 |
| J2K adaptive (1/24), clean      | 9.9                | 9.9 | 9.9 | 9.9 | 9.9  | 9.9  | 13.5 | 25.2 | 25.2 | 44.7 | 48.6 | 48.6 | 59.2 | 59.2 | 75.1 | 79.8 | 80.0 | 84.4 | 85.9 | 85.9 | 85.9 |

TABLE S12

PAIRED TEST-ACCURACY DIFFERENCE, DJSCC MINUS EACH DIGITAL SYSTEM, AT  $r = 1/6$ , IN POINTS, AVERAGED OVER THREE SEED PAIRS WITHIN EACH IMAGE, WITH 95% IMAGE-BOOTSTRAP INTERVALS (10,000 RESAMPLES, SHARED WITH THE HYPOTHESIS TESTS).

| SNR (dB) | vs. J2K adaptive |                |  | vs. task-aware digital |                |  |
|----------|------------------|----------------|--|------------------------|----------------|--|
|          | $\Delta$         | 95% CI         |  | $\Delta$               | 95% CI         |  |
| -8       | +66.3            | [+64.8, +67.8] |  | +66.3                  | [+64.8, +67.8] |  |
| -7       | +67.6            | [+66.2, +69.1] |  | +67.6                  | [+66.2, +69.1] |  |
| -6       | +69.2            | [+67.8, +70.7] |  | +69.2                  | [+67.8, +70.7] |  |
| -5       | +70.2            | [+68.8, +71.6] |  | +70.2                  | [+68.8, +71.6] |  |
| -4       | -0.9             | [-2.0, +0.3]   |  | -0.3                   | [-1.0, +0.4]   |  |
| -3       | -0.7             | [-1.9, +0.4]   |  | -0.2                   | [-0.9, +0.5]   |  |
| -2       | -1.8             | [-2.9, -0.7]   |  | +0.3                   | [-0.4, +0.9]   |  |
| -1       | -4.2             | [-5.2, -3.1]   |  | +0.6                   | [-0.0, +1.3]   |  |
| 0        | -4.3             | [-5.4, -3.3]   |  | +0.5                   | [-0.2, +1.1]   |  |
| 1        | -4.1             | [-5.1, -3.1]   |  | +0.7                   | [+0.1, +1.4]   |  |
| 2        | -4.2             | [-5.2, -3.2]   |  | +0.9                   | [+0.2, +1.5]   |  |
| 3        | -4.0             | [-5.0, -3.0]   |  | +1.1                   | [+0.4, +1.7]   |  |
| 4        | -4.0             | [-5.0, -3.0]   |  | +1.0                   | [+0.4, +1.7]   |  |
| 5        | -3.8             | [-4.8, -2.8]   |  | +1.3                   | [+0.6, +1.9]   |  |
| 6        | -4.3             | [-5.3, -3.3]   |  | +1.1                   | [+0.5, +1.8]   |  |
| 7        | -4.2             | [-5.2, -3.2]   |  | +1.2                   | [+0.6, +1.9]   |  |
| 9        | -2.0             | [-3.1, -1.0]   |  | +1.2                   | [+0.5, +1.8]   |  |
| 11       | -4.2             | [-5.2, -3.2]   |  | +1.3                   | [+0.6, +1.9]   |  |
| 13       | -4.2             | [-5.2, -3.2]   |  | +1.3                   | [+0.6, +1.9]   |  |
| 15       | -4.2             | [-5.2, -3.1]   |  | +1.3                   | [+0.7, +2.0]   |  |
| 18       | -4.2             | [-5.2, -3.2]   |  | +1.3                   | [+0.7, +2.0]   |  |

TABLE S13

POINT STATISTICS FOR H1 (DJSCC VS. ADAPTIVE JPEG 2000) AND H4 (DJSCC VS. TASK-AWARE DIGITAL) OVER THE REGION AT OR BELOW 7 dB: MEAN PAIRED DIFFERENCE (POINTS), STUDENTIZED STATISTIC  $t$ , AND WHETHER THE POINT QUALIFIES ( $t > 1.96$ ). H1: LONGEST RUN 4, CALIBRATED  $p = 0.0303$ . H4: LONGEST RUN 7, CALIBRATED  $p = 0.0096$ .

| SNR (dB) | H1       |      |   | H4       |      |   |
|----------|----------|------|---|----------|------|---|
|          | $\Delta$ | $t$  | Q | $\Delta$ | $t$  | Q |
| -8       | +66.3    | 87.8 | ✓ | +66.3    | 87.8 | ✓ |
| -7       | +67.6    | 91.0 | ✓ | +67.6    | 91.0 | ✓ |
| -6       | +69.2    | 94.4 | ✓ | +69.2    | 94.4 | ✓ |
| -5       | +70.2    | 97.5 | ✓ | +70.2    | 97.5 | ✓ |
| -4       | -0.9     | -1.5 |   | -0.3     | -1.0 |   |
| -3       | -0.7     | -1.3 |   | -0.2     | -0.6 |   |
| -2       | -1.8     | -3.3 |   | +0.3     | 0.9  |   |
| -1       | -4.2     | -7.9 |   | +0.6     | 1.9  |   |
| 0        | -4.3     | -8.2 |   | +0.5     | 1.4  |   |
| 1        | -4.1     | -7.8 |   | +0.7     | 2.1  | ✓ |
| 2        | -4.2     | -8.1 |   | +0.9     | 2.6  | ✓ |
| 3        | -4.0     | -7.7 |   | +1.1     | 3.2  | ✓ |
| 4        | -4.0     | -7.8 |   | +1.0     | 3.1  | ✓ |
| 5        | -3.8     | -7.4 |   | +1.3     | 3.8  | ✓ |
| 6        | -4.3     | -8.4 |   | +1.1     | 3.4  | ✓ |
| 7        | -4.2     | -8.2 |   | +1.2     | 3.8  | ✓ |

TABLE S14

OPERATING POINTS OF THE ADAPTIVE JPEG 2000 BASELINE AT  $r = 1/6$ , SELECTED ON VALIDATION DATA, WITH TEST ACCURACY AND DELIVERED FRACTION.

| SNR (dB) | Image (px) | LDPC | Mod.   | Acc. (%) | Deliv. (%) |
|----------|------------|------|--------|----------|------------|
| -8       | 160        | 1/3  | BPSK   | 9.9      | 0.0        |
| -7       | 160        | 1/3  | BPSK   | 9.9      | 0.0        |
| -6       | 160        | 1/3  | BPSK   | 9.9      | 0.0        |
| -5       | 160        | 1/3  | BPSK   | 9.9      | 0.0        |
| -4       | 64         | 1/3  | BPSK   | 81.6     | 100.0      |
| -3       | 64         | 1/3  | BPSK   | 81.6     | 100.0      |
| -2       | 96         | 1/2  | BPSK   | 83.2     | 97.6       |
| -1       | 128        | 1/3  | QPSK   | 85.9     | 100.0      |
| 0        | 128        | 1/3  | QPSK   | 85.9     | 100.0      |
| 1        | 128        | 2/3  | BPSK   | 85.9     | 100.0      |
| 2        | 128        | 1/2  | QPSK   | 86.2     | 100.0      |
| 3        | 128        | 1/2  | QPSK   | 86.2     | 100.0      |
| 4        | 128        | 1/2  | QPSK   | 86.2     | 100.0      |
| 5        | 128        | 1/2  | QPSK   | 86.2     | 100.0      |
| 6        | 160        | 5/6  | QPSK   | 86.5     | 100.0      |
| 7        | 160        | 1/2  | 16-QAM | 86.5     | 100.0      |
| 9        | 160        | 2/3  | 16-QAM | 84.3     | 96.6       |
| 11       | 160        | 2/3  | 16-QAM | 86.6     | 100.0      |
| 13       | 160        | 2/3  | 16-QAM | 86.6     | 100.0      |
| 15       | 160        | 2/3  | 16-QAM | 86.6     | 100.0      |
| 18       | 160        | 2/3  | 16-QAM | 86.6     | 100.0      |

TABLE S15

OPERATING POINTS OF THE ADAPTIVE JPEG 2000 BASELINE AT  $r = 1/24$ , SELECTED ON VALIDATION DATA, WITH TEST ACCURACY AND DELIVERED FRACTION.

| SNR (dB) | Image (px) | LDPC | Mod.   | Acc. (%) | Deliv. (%) |
|----------|------------|------|--------|----------|------------|
| -8       | 160        | 1/3  | BPSK   | 9.9      | 0.0        |
| -7       | 160        | 1/3  | BPSK   | 9.9      | 0.0        |
| -6       | 160        | 1/3  | BPSK   | 9.9      | 0.0        |
| -5       | 160        | 1/3  | BPSK   | 9.9      | 0.0        |
| -4       | 160        | 1/3  | BPSK   | 9.9      | 0.0        |
| -3       | 160        | 1/3  | BPSK   | 9.9      | 0.0        |
| -2       | 96         | 1/2  | BPSK   | 28.9     | 82.0       |
| -1       | 160        | 1/3  | QPSK   | 67.9     | 99.9       |
| 0        | 160        | 1/3  | QPSK   | 68.0     | 100.0      |
| 1        | 96         | 1/2  | QPSK   | 72.6     | 91.2       |
| 2        | 96         | 1/2  | QPSK   | 79.1     | 100.0      |
| 3        | 96         | 1/2  | QPSK   | 79.1     | 100.0      |
| 4        | 64         | 1/3  | 16-QAM | 81.6     | 100.0      |
| 5        | 64         | 1/3  | 16-QAM | 81.6     | 100.0      |
| 6        | 96         | 5/6  | QPSK   | 83.9     | 100.0      |
| 7        | 96         | 1/2  | 16-QAM | 84.9     | 100.0      |
| 9        | 128        | 2/3  | 16-QAM | 81.5     | 94.5       |
| 11       | 128        | 2/3  | 16-QAM | 85.9     | 100.0      |
| 13       | 128        | 5/6  | 16-QAM | 86.3     | 100.0      |
| 15       | 128        | 5/6  | 16-QAM | 86.3     | 100.0      |
| 18       | 128        | 5/6  | 16-QAM | 86.3     | 100.0      |

TABLE S16

OPERATING POINTS OF THE BASELINE JPEG SYSTEM AT  $r = 1/6$ ,  
SELECTED ON VALIDATION DATA, WITH TEST ACCURACY AND DELIVERED  
FRACTION.

| SNR (dB) | Image (px) | JPEG $Q$ | LDPC | Mod.   | Acc. (%) | Deliv. (%) |
|----------|------------|----------|------|--------|----------|------------|
| -8       | 64         | 5        | 1/3  | BPSK   | 9.9      | 0.0        |
| -7       | 64         | 5        | 1/3  | BPSK   | 9.9      | 0.0        |
| -6       | 64         | 5        | 1/3  | BPSK   | 9.9      | 0.0        |
| -5       | 64         | 5        | 1/3  | BPSK   | 9.9      | 0.0        |
| -4       | 64         | 5        | 1/3  | BPSK   | 66.4     | 99.7       |
| -3       | 64         | 5        | 1/3  | BPSK   | 66.4     | 99.7       |
| -2       | 64         | 10       | 1/2  | BPSK   | 74.5     | 97.8       |
| -1       | 64         | 20       | 1/3  | QPSK   | 81.2     | 100.0      |
| 0        | 64         | 20       | 1/3  | QPSK   | 81.2     | 100.0      |
| 1        | 64         | 20       | 2/3  | BPSK   | 81.2     | 100.0      |
| 2        | 96         | 15       | 1/2  | QPSK   | 84.3     | 100.0      |
| 3        | 96         | 15       | 1/2  | QPSK   | 84.3     | 100.0      |
| 4        | 96         | 30       | 2/3  | QPSK   | 85.1     | 100.0      |
| 5        | 96         | 30       | 2/3  | QPSK   | 85.1     | 100.0      |
| 6        | 128        | 20       | 5/6  | QPSK   | 85.5     | 100.0      |
| 7        | 128        | 20       | 5/6  | QPSK   | 85.5     | 100.0      |
| 9        | 160        | 15       | 2/3  | 16-QAM | 9.9      | 0.0        |
| 11       | 160        | 15       | 2/3  | 16-QAM | 86.1     | 100.0      |
| 13       | 160        | 15       | 2/3  | 16-QAM | 86.1     | 100.0      |
| 15       | 160        | 15       | 2/3  | 16-QAM | 86.1     | 100.0      |
| 18       | 160        | 15       | 2/3  | 16-QAM | 86.1     | 100.0      |

TABLE S17

OPERATING POINTS OF THE QPSK-ONLY JPEG 2000 BASELINE AT  
 $r = 1/6$ , SELECTED ON VALIDATION DATA, WITH TEST ACCURACY AND  
DELIVERED FRACTION.

| SNR (dB) | Image (px) | LDPC | Mod. | Acc. (%) | Deliv. (%) |
|----------|------------|------|------|----------|------------|
| -8       | 160        | 1/3  | QPSK | 9.9      | 0.0        |
| -7       | 160        | 1/3  | QPSK | 9.9      | 0.0        |
| -6       | 160        | 1/3  | QPSK | 9.9      | 0.0        |
| -5       | 160        | 1/3  | QPSK | 9.9      | 0.0        |
| -4       | 160        | 1/3  | QPSK | 9.9      | 0.0        |
| -3       | 160        | 1/3  | QPSK | 9.9      | 0.0        |
| -2       | 160        | 1/3  | QPSK | 9.9      | 0.0        |
| -1       | 128        | 1/3  | QPSK | 85.9     | 100.0      |
| 0        | 128        | 1/3  | QPSK | 85.9     | 100.0      |
| 1        | 128        | 1/3  | QPSK | 85.9     | 100.0      |
| 2        | 128        | 1/2  | QPSK | 86.2     | 100.0      |
| 3        | 128        | 1/2  | QPSK | 86.2     | 100.0      |
| 4        | 128        | 1/2  | QPSK | 86.2     | 100.0      |
| 5        | 128        | 1/2  | QPSK | 86.2     | 100.0      |
| 6        | 160        | 5/6  | QPSK | 86.5     | 100.0      |
| 7        | 160        | 5/6  | QPSK | 86.5     | 100.0      |
| 9        | 160        | 5/6  | QPSK | 86.5     | 100.0      |
| 11       | 160        | 5/6  | QPSK | 86.5     | 100.0      |
| 13       | 160        | 5/6  | QPSK | 86.5     | 100.0      |
| 15       | 160        | 5/6  | QPSK | 86.5     | 100.0      |
| 18       | 160        | 5/6  | QPSK | 86.5     | 100.0      |

## S6. ADDITIONAL TEST FIGURES

Figs. S1–S4 plot four comparisons that the paper reports in its text and in its Table I: the two bandwidth ratios, the two receiver classifiers, the three DJSCC training recipes, and the restricted digital baselines. Scorers and operating-point selection are as in Fig. 1 of the paper.

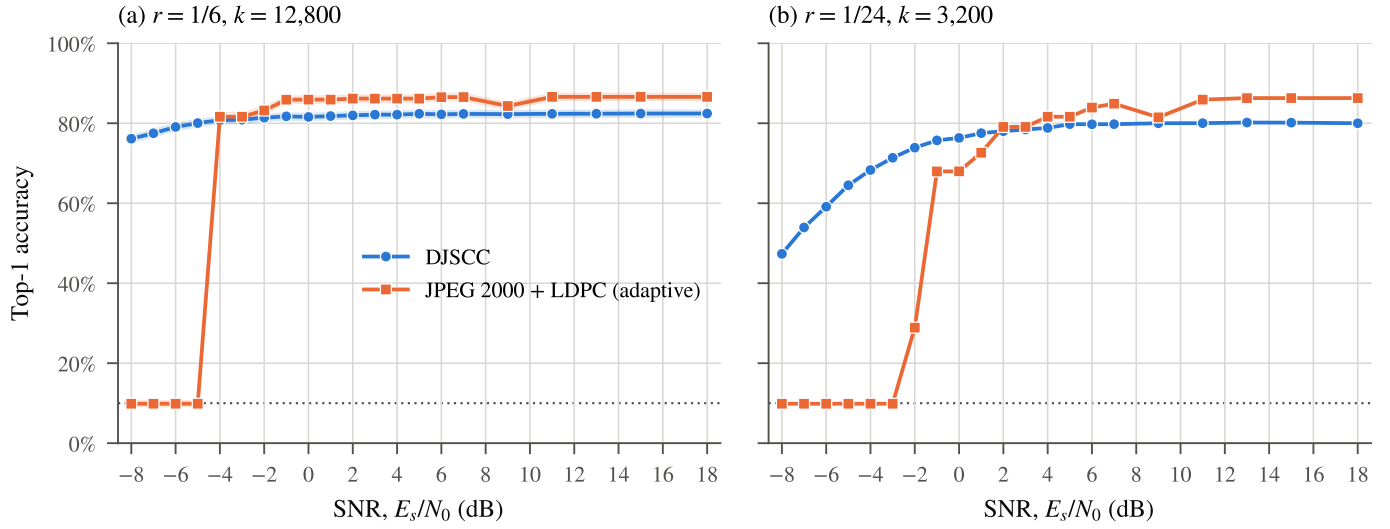


Fig. S1. DJSCC against the adaptive JPEG 2000 baseline on the test split at two bandwidth ratios. (a) Three seed pairs with 95% intervals; (b) the first seed pair. Reducing the channel uses fourfold moves the first SNR at which the digital system is ahead from  $-2$  dB to  $+2$  dB.

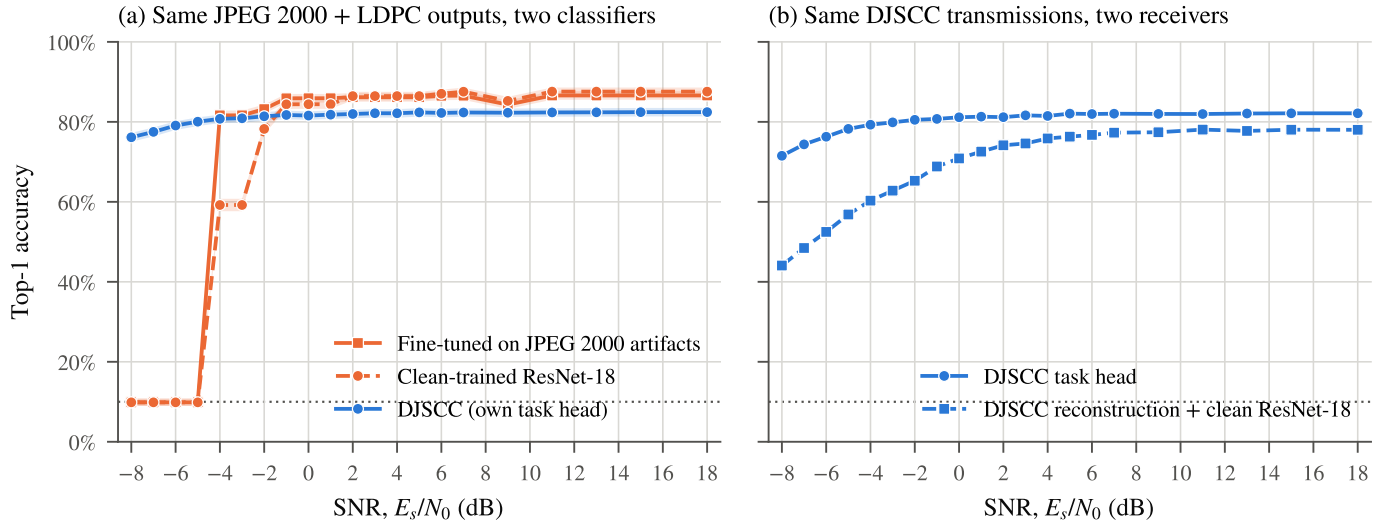


Fig. S2. (a) The same JPEG 2000 + LDPC outputs at  $r = 1/6$  scored by the fine-tuned and the clean ResNet-18, with DJSCC for reference (three seed pairs). (b) The same DJSCC transmissions read by the DJSCC task head and by the clean ResNet-18 applied to the DJSCC reconstruction (first seed pair).

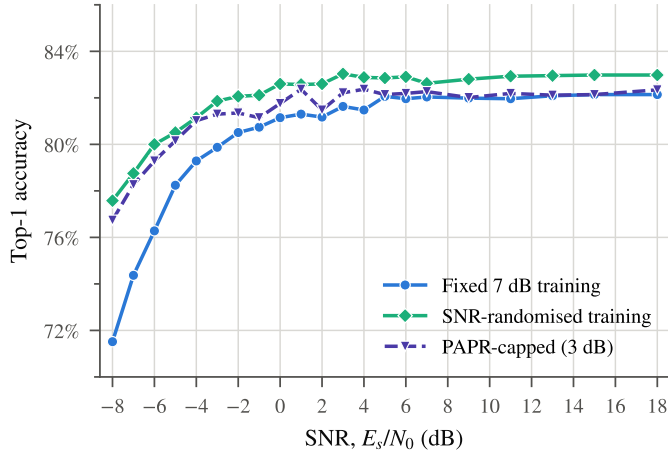


Fig. S3. DJSCC at  $r = 1/6$  on the test split, first seed pair: trained at a fixed 7 dB, trained with SNR drawn from  $\{1, 4, 7, 13, 19\}$  dB, and trained with a 3 dB symbol PAPR cap. The other two seed pairs of the fixed-SNR model score 78.4% and 78.5% at  $-8$  dB. Note the vertical scale.

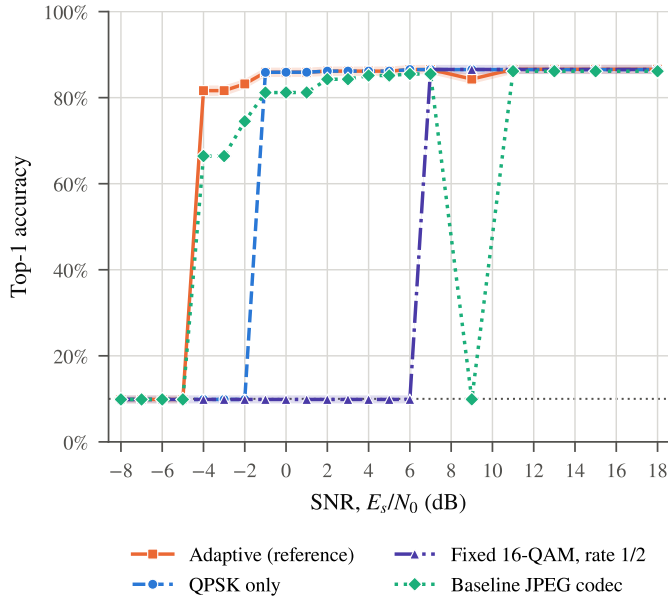


Fig. S4. Adaptive JPEG 2000 + LDPC at  $r = 1/6$  on the test split against restricted versions: QPSK only (rate and resolution still adapted), one fixed operating point at all SNRs, and baseline JPEG in place of JPEG 2000. All scored by the fine-tuned ResNet-18.

## S7. REPRODUCING THE FIGURES AND TABLES

The figures and tables in the paper and in this document are regenerated from the committed result files, without model weights, datasets, or a GPU:

---

```
git clone https://github.com/nxck2005/capstone
cd capstone
python3 deliverables/research-paper/figures/make_figures.py
python3 deliverables/research-paper/supplement/make_tables.py
cd deliverables/research-paper
latexmk -pdf capstone_rp.tex
cd supplement && latexmk -pdf capstone_rp_supplement.tex
```

---

Both scripts read only committed files: presentation-results/data/g12\_test\_\*.csv, results/g12/, and results/learned/w10/w10\_continuation\_units\_v11.json. The two CSV tables are written by tools/export\_g12\_tables.py from the per-image test outcomes, which are archived outside the repository and bound by the digests in results/per\_image\_manifest.csv. Re-running training or evaluation requires the Imagenette-160 archive, an NVIDIA GPU, and the environment pinned in requirements.lock.

The test campaign itself ran once under the committed freeze manifest results/freeze\_manifest.json. Its execution record is in results/g12/: attempt1\_abort\_custody.json documents a first launch that stopped before any model produced a test output, and post\_freeze\_corrections/ records, with the exact files and contents changed, the two execution corrections described in Section V-A of the paper. Neither changed a model, an operating point, a noise draw, an analysis rule, or a reported accuracy.

### ACKNOWLEDGMENT

This supplementary material was produced with Anthropic’s Claude Opus 5.5. It wrote the pseudo-code in Section S1 from the project’s source code, selected and condensed the code excerpts in Section S2, drafted the text throughout, and generated the tables in Sections S3–S5 and the figures in Section S6 with scripts that read the published result files. The code excerpted in Section S2 was written with the assistance of Claude Opus 5.5 and OpenAI’s GPT-5.6 Sol. All content was reviewed and verified by the authors, who take full responsibility for it.